E-ASSET MANAGEMENT SYSTEM

E-AMS

Table of Contents

1. Executive Summary

The E-Asset Management System (E-AMS) is a fully customisable, AI-integrated enterprise software platform designed to serve as a single authoritative source of truth for all IT assets across any organisation — regardless of industry, size, or geographic distribution. Unlike purpose-built tools tied to a specific institution, E-AMS is architected around a generic, configurable data model that any company can tailor to its own asset taxonomy, organisational hierarchy, nomenclature, and compliance requirements without modifying the underlying codebase.

The system addresses six core operational domains: hardware and software asset tracking, infrastructure and network device management, security credential management (certificates and software licenses), governance and compliance, AI-powered predictive analytics, and configurable workflow automation. E-AMS replaces fragmented spreadsheets and disconnected vendor portals with a structured relational database, enforces segregation of duties through a role-based access control model, maintains a tamper-evident audit trail of every action, and surfaces actionable intelligence through an embedded AI analytics engine.

This document constitutes the formal Software Requirements Specification (SRS) for E-AMS version 2.0. It defines all functional and non-functional requirements, presents the complete database schema for every module, and serves as the contractual baseline between the development team, procurement stakeholders, and system administrators.

2. Design Principles and Scope

2.1 Problem Statement

Most organisations — whether financial institutions, healthcare providers, government agencies, or commercial enterprises — manage IT assets through a combination of spreadsheets, departmental registers, and fragmented vendor portals. This creates systemic risks: inventories become stale, software licence compliance cannot be demonstrated, security certificates expire without warning, temporary access grants are never revoked, and none of these events are captured in a tamper-evident audit trail. No single commercial product addresses all of these domains in a configurable, institution-agnostic manner.

2.2 Customisation Philosophy

E-AMS is built on a “configure, don’t code” principle. Every aspect of the system that varies between organisations — asset categories, field labels, workflow steps, role names, notification recipients, report templates, and AI model thresholds — is controlled through the Administration Console without requiring source code changes. This allows the same deployed instance to serve a bank, a hospital, a university, or a government ministry with equal fidelity.

2.3 Scope

E-AMS covers all IT assets owned, leased, or operated by the organisation across all physical locations, cloud environments, and third-party hosted services. The default asset taxonomy ships with six pre-configured asset categories — software applications, databases, servers and virtualisation infrastructure, network devices, client devices, and security credentials — but administrators may add, rename, or remove categories. The system also manages the people dimension (users, roles, organisational hierarchy) and transactional events such as asset check-in/out, change requests, and access grants.

3. Stakeholders and User Roles

3.1 Generic Stakeholder Roles

The following roles represent generic organisational archetypes. Role names, titles, and permission sets are fully configurable in the Administration Console to match any organisation's actual structure.

Stakeholder / Role	Function in the System	Primary Concern
Executive / C-Level	Strategic oversight; approves major changes; reviews executive dashboards	Total asset value, compliance posture, risk exposure
IT Security Officer	Security governance; reviews certificate, access, and vulnerability data	Certificate health, access control, audit findings
Head of IT / IT Director	Day-to-day platform administration; asset lifecycle decisions	Asset accuracy, lifecycle status, capacity planning
Systems / Network Administrator	Asset data entry, configuration management, technical review	Device details, software versions, topology
IT Asset Manager	Licence compliance, procurement tracking, assignment records	Licence utilisation, check-in/out, contract expiry
Internal Auditor	Compliance evidence, access reviews, policy adherence	Audit trails, change logs, segregation of duties
External Auditor / Inspector	Independent verification of IT controls and asset inventories	Exportable reports, complete audit evidence
Procurement Officer	Contract management, renewal coordination, vendor relationships	Costs, expiry dates, purchase order references
Branch / Regional IT Staff	Local asset registration, device assignment, basic helpdesk	Easy data entry, localised asset view
Finance Department	Budget planning, depreciation tracking, licence cost management	Asset cost, depreciation, total licence spend
AI / Data Analytics Team	Model tuning, anomaly threshold adjustment, insight review	Prediction accuracy, data quality, model drift

3.2 System Roles and Segregation of Duties

Roles and permission sets are defined in the Administration Console. The following defaults ship with the system and are recommended as a starting baseline.

Role	Asset Register	Approve Changes	Grant Access	Audit / Report	Description
System Admin	Full CRUD	Yes	No	Read only	Full technical access; cannot modify audit trail

Role	Asset Register	Approve Changes	Grant Access	Audit / Report	Description
Asset Manager	Create / Edit	No	No	Read only	Manages inventory records and assignments only
Security Officer	Read / Review	No	Yes	Full	Manages access grants and certificates; cannot approve own grants
Change Manager	Read only	Yes	No	Read only	Approves changes; cannot also be the implementer
Auditor	Read only	No	No	Full export	Read-only access to all data; full audit log export
Branch Staff	Own branch only	No	No	No	Registers and views assets in their branch only
Procurement Officer	Cost fields only	No	No	Read only	Views and updates contract, cost, and licence data
AI Analyst	Read only	No	No	AI reports	Accesses AI dashboards, model outputs, and recommendations
Read-Only Viewer	Read only	No	No	No	Dashboard viewing only — no data modification rights

4. Core Database Schema

All table and column names below are the physical database names. Every table includes standard system columns: id (UUID PK), created_at (TIMESTAMPTZ), updated_at (TIMESTAMPTZ), created_by (UUID FK → users.id), and is_deleted (BOOLEAN DEFAULT FALSE). These are omitted from individual schemas below for clarity but are mandatory in all tables.

4.1 organisations Table

Supports multi-tenant deployments. Every other table references organisations.id, enabling complete data isolation between tenants.

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique organisation identifier
name	VARCHAR(255)	NOT NULL	Full legal name of the organisation
slug	VARCHAR(100)	UNIQUE, NOT NULL	URL-safe short identifier for the organisation
logo_url	TEXT	NULLABLE	URL to the organisation's logo for white-label branding
primary_color	VARCHAR(7)	DEFAULT #1B3A6B	Hex colour for UI theming
timezone	VARCHAR(64)	NOT NULL	Default timezone (e.g., Africa/Kigali, Europe/London)

Column Name	Data Type	Constraints	Description
locale	VARCHAR(10)	DEFAULT en	Default locale for date/number formatting
country_code	CHAR(2)	NOT NULL	ISO 3166-1 alpha-2 country code
subscription_plan	VARCHAR(50)	DEFAULT standard	Licensing tier: trial / standard / enterprise
max_assets	INTEGER	DEFAULT 10000	Maximum asset records allowed under current plan
is_active	BOOLEAN	DEFAULT TRUE	Whether this tenant is active
settings	JSONB	NULLABLE	Key-value store for organisation-level configuration overrides

4.2 locations Table

Stores the full organisational hierarchy: Organisation → Country → Region → Branch → Building → Floor → Room → Rack. Assets are assigned to any level of this hierarchy.

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique location identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
parent_id	UUID	FK → locations.id, NULLABLE	Parent location (NULL for root)
location_type	VARCHAR(50)	NOT NULL	Enum: country / region / branch / building / floor / room / rack
name	VARCHAR(255)	NOT NULL	Display name (e.g., 'Nairobi Branch', 'Server Room A')
code	VARCHAR(20)	NOT NULL	Short code used in Asset ID generation (e.g., NBI)
address	TEXT	NULLABLE	Physical address (for branches and above)
latitude	DECIMAL(10,7)	NULLABLE	GPS latitude for map-based asset views
longitude	DECIMAL(10,7)	NULLABLE	GPS longitude for map-based asset views
contact_email	VARCHAR(255)	NULLABLE	Primary contact email for this location
is_active	BOOLEAN	DEFAULT TRUE	Whether this location is currently operational

4.3 departments Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique department identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
location_id	UUID	FK → locations.id, NULLABLE	Primary branch/location of the department
parent_id	UUID	FK → departments.id, NULLABLE	Parent department for hierarchical structures
name	VARCHAR(255)	NOT NULL	Department name (e.g., ICT Infrastructure, Finance)
code	VARCHAR(20)	UNIQUE, NOT NULL	Short code used in reporting
head_user_id	UUID	FK → users.id, NULLABLE	Department head / manager
cost_centre	VARCHAR(50)	NULLABLE	Financial cost centre code for budget allocation
is_active	BOOLEAN	DEFAULT TRUE	Whether the department is active

4.4 users Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique user identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
employee_id	VARCHAR(50)	UNIQUE PER ORG, NOT NULL	Organisation's internal employee / staff number
full_name	VARCHAR(255)	NOT NULL	Full legal name
email	VARCHAR(255)	UNIQUE, NOT NULL	Primary email address (also login identifier)
phone	VARCHAR(30)	NULLABLE	Mobile number for SMS notifications
job_title	VARCHAR(150)	NULLABLE	Job title / designation
department_id	UUID	FK → departments.id, NULLABLE	Department the user belongs to
location_id	UUID	FK → locations.id, NULLABLE	Primary branch / work location
line_manager_id	UUID	FK → users.id, NULLABLE	Direct line manager (self-referencing)
ad_username	VARCHAR(255)	NULLABLE	Active Directory / LDAP username for SSO sync
employment_type	VARCHAR(30)	DEFAULT permanent	Enum: permanent / contractor / vendor / intern

Column Name	Data Type	Constraints	Description
employment_status	VARCHAR(30)	DEFAULT active	Enum: active / suspended / terminated
access_expiry_date	DATE	NULLABLE	For contractors/vendors: mandatory access expiry date
mfa_enabled	BOOLEAN	DEFAULT FALSE	Whether MFA is currently active for this user
mfa_method	VARCHAR(30)	NULLABLE	Enum: totp / fido2 / sms
last_login_at	TIMESTAMPTZ	NULLABLE	Timestamp of most recent successful login
failed_login_count	SMALLINT	DEFAULT 0	Consecutive failed login attempts (resets on success)
locked_at	TIMESTAMPTZ	NULLABLE	Timestamp when account was locked due to failed logins
password_hash	TEXT	NULLABLE	Bcrypt hash (NULL when using SSO-only authentication)
password_changed_at	TIMESTAMPTZ	NULLABLE	Timestamp of last password change
avatar_url	TEXT	NULLABLE	Profile photo URL
is_active	BOOLEAN	DEFAULT TRUE	Master active flag

4.5 roles & role_permissions Tables

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique role identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Tenant that owns this role
name	VARCHAR(100)	NOT NULL	Display name (e.g., ‘System Admin’, ‘Asset Manager’)
code	VARCHAR(50)	NOT NULL	System code (used in permission checks)
description	TEXT	NULLABLE	Plain-language description of this role’s purpose
is_system_role	BOOLEAN	DEFAULT FALSE	System roles cannot be deleted; only renamed
requires_mfa	BOOLEAN	DEFAULT TRUE	Whether users in this role must have MFA active
session_timeout_minutes	INTEGER	DEFAULT 30	Inactivity timeout for sessions under this role

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Row identifier
role_id	UUID	FK → roles.id, NOT NULL	The role being granted this permission
permission_code	VARCHAR(100)	NOT NULL	Machine-readable permission code (e.g., asset.create, report.export)
asset_category_id	UUID	FK → asset_categories.id, NULLABLE	Scope permission to a specific asset category (NULL = all)
location_id	UUID	FK → locations.id, NULLABLE	Scope permission to a specific location (NULL = all)
can_create	BOOLEAN	DEFAULT FALSE	Create new records
can_read	BOOLEAN	DEFAULT FALSE	View records
can_update	BOOLEAN	DEFAULT FALSE	Edit existing records
can_delete	BOOLEAN	DEFAULT FALSE	Soft-delete records
can_approve	BOOLEAN	DEFAULT FALSE	Approve pending workflow actions
can_export	BOOLEAN	DEFAULT FALSE	Export data to file formats

4.6 asset_categories Table

> *This table drives the dynamic asset type system. Administrators define custom asset types here without code changes.*

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique category identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Tenant that owns this category
name	VARCHAR(100)	NOT NULL	Display name (e.g., 'Software Application', 'Network Device')
code	VARCHAR(20)	NOT NULL	Short code used in Asset ID prefix (e.g., APP, NET, SRV)
icon	VARCHAR(50)	NULLABLE	Icon identifier for UI display
description	TEXT	NULLABLE	Description of what assets belong to this category
id_format	VARCHAR(100)	NOT NULL	Template for Asset ID generation (e.g., {CODE}-{LOC}-{YYYY}-{SEQ:4})
default_criticality	VARCHAR(20)	DEFAULT medium	Default criticality for new assets: critical / high / medium / low
depreciation_years	INTEGER	NULLABLE	Default useful life in years for depreciation calculation

Column Name	Data Type	Constraints	Description
is_physical	BOOLEAN	DEFAULT FALSE	Whether assets in this category have a physical form factor
requires_location	BOOLEAN	DEFAULT TRUE	Whether a physical location assignment is mandatory
is_active	BOOLEAN	DEFAULT TRUE	Whether this category accepts new asset registrations
sort_order	INTEGER	DEFAULT 0	Display order in the UI

4.7 custom_fields Table

> *Organisations may add unlimited custom fields to any asset category without schema changes. Field values are stored in asset_custom_values.*

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique field identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
asset_category_id	UUID	FK → asset_categories.id, NOT NULL	Category this field belongs to
field_label	VARCHAR(150)	NOT NULL	Human-readable label shown in the UI
field_name	VARCHAR(100)	NOT NULL	Machine-readable key (snake_case)
field_type	VARCHAR(30)	NOT NULL	Enum: text / number / date / dropdown / boolean / url / email / file
options	JSONB	NULLABLE	For dropdown fields: array of allowed option values
is_mandatory	BOOLEAN	DEFAULT FALSE	Whether this field must be filled before save
is_searchable	BOOLEAN	DEFAULT TRUE	Whether this field is indexed for search
default_value	TEXT	NULLABLE	Default value pre-populated on new asset forms
validation_regex	TEXT	NULLABLE	Optional regex for field-level validation
help_text	TEXT	NULLABLE	Tooltip / contextual help shown beside the field
sort_order	INTEGER	DEFAULT 0	Display order within the asset form
is_active	BOOLEAN	DEFAULT TRUE	Whether this field is currently shown

4.8 assets Table (Core)

> *The central asset record. All six asset categories share this table. Category-specific fields are stored in asset_extended_{category} tables (see Section 6) and configurable fields in asset_custom_values.*

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique asset identifier (system-wide)
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
asset_id_display	VARCHAR(60)	UNIQUE PER ORG, NOT NULL	Human-readable Asset ID (e.g., APP-NBI-2026-0042)
asset_category_id	UUID	FK → asset_categories.id, NOT NULL	Category / type of asset
name	VARCHAR(255)	NOT NULL	Primary name or identifier of the asset
description	TEXT	NULLABLE	Free-text description of the asset
location_id	UUID	FK → locations.id, NULLABLE	Physical or logical location
department_id	UUID	FK → departments.id, NULLABLE	Owning department
assigned_user_id	UUID	FK → users.id, NULLABLE	Currently assigned user (for client devices)
technical_owner_id	UUID	FK → users.id, NULLABLE	Primary technical responsible person
business_owner_id	UUID	FK → users.id, NULLABLE	Business owner / process owner
backup_owner_id	UUID	FK → users.id, NULLABLE	Secondary / backup responsible person
criticality	VARCHAR(20)	NOT NULL DEFAULT medium	Enum: critical / high / medium / low
status	VARCHAR(30)	NOT NULL DEFAULT active	Enum: active / inactive / decommissioned / in_maintenance / in_stock / lost / stolen
year_of_installation	SMALLINT	NULLABLE	Year asset was placed into service
acquisition_cost	NUMERIC(15,2)	NULLABLE	Purchase / acquisition cost in base currency
currency_code	CHAR(3)	DEFAULT USD	ISO 4217 currency code
annual_cost	NUMERIC(15,2)	NULLABLE	Ongoing annual cost (subscription, maintenance)
purchase_order_ref	VARCHAR(100)	NULLABLE	PO or contract reference number
vendor_name	VARCHAR(255)	NULLABLE	Primary vendor or manufacturer name
support_contract_ref	VARCHAR(100)	NULLABLE	Vendor support / maintenance contract reference
support_expiry_date	DATE	NULLABLE	Date vendor support ends
warranty_expiry_date	DATE	NULLABLE	Hardware warranty expiry date

Column Name	Data Type	Constraints	Description
last_audit_date	DATE	NULLABLE	Date of last formal record verification
next_audit_due_date	DATE	NULLABLE	Calculated next audit date based on criticality schedule
notes	TEXT	NULLABLE	Free-text additional context
qr_code_url	TEXT	NULLABLE	URL to the generated QR code for physical labelling
barcode	VARCHAR(100)	NULLABLE	Barcode value for physical scanning
is_deleted	BOOLEAN	DEFAULT FALSE	Soft-delete flag
lifecycle_stage	VARCHAR(50)	DEFAULT in_service	Enum: in_procurement / in_stock / in_service / in_maintenance / decommissioned
decommission_date	DATE	NULLABLE	Date asset was formally decommissioned
decommission_reason	TEXT	NULLABLE	Reason for decommissioning

4.9 asset_relationships Table

CMDB-style relationships between assets. Supports directed typed relationships to model dependencies, hosting, and connectivity.

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Relationship record identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
source_asset_id	UUID	FK → assets.id, NOT NULL	The asset that 'owns' or initiates the relationship
target_asset_id	UUID	FK → assets.id, NOT NULL	The related asset
relationship_type	VARCHAR(50)	NOT NULL	Enum: runs_on / depends_on / is_component_of / is_backed_up_by / connects_to / hosts / secures
description	TEXT	NULLABLE	Optional description of the relationship
is_active	BOOLEAN	DEFAULT TRUE	Whether this relationship is currently valid

4.10 audit_logs Table

Immutable append-only audit trail. No UPDATE or DELETE is permitted on this table at the database level (enforced via table privileges and triggers).

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Unique log entry identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
event_timestamp	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	UTC timestamp of the event
actor_user_id	UUID	FK → users.id, NULLABLE	User who performed the action (NULL for system events)
actor_display_name	VARCHAR(255)	NOT NULL	Denormalised user name at time of action
actor_ip_address	INET	NULLABLE	IP address of the actor's session
actor_user_agent	TEXT	NULLABLE	Browser or API client user-agent string
session_id	UUID	NULLABLE	Session that generated this event
action_type	VARCHAR(50)	NOT NULL	Enum: create / read / update / delete / login / logout / export / approve / reject / assign / revoke
resource_type	VARCHAR(100)	NOT NULL	Type of entity affected (e.g., asset, user, certificate, workflow_step)
resource_id	UUID	NULLABLE	UUID of the affected record
resource_display	VARCHAR(255)	NULLABLE	Human-readable name of the affected resource at time of action
field_name	VARCHAR(150)	NULLABLE	For updates: the field that was changed
old_value	TEXT	NULLABLE	Previous value before the change
new_value	TEXT	NULLABLE	New value after the change
change_reason	TEXT	NULLABLE	Optional business justification provided by the actor
signature	TEXT	NOT NULL	HMAC-SHA256 signature for tamper detection
is_sensitive	BOOLEAN	DEFAULT FALSE	Whether this log entry contains sensitive data (redacted in UI for non-auditors)

4.11 check_in_out_transactions Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Transaction identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
asset_id	UUID	FK → assets.id, NOT NULL	The asset being transacted

Column Name	Data Type	Constraints	Description
transaction_type	VARCHAR(20)	NOT NULL	Enum: check_out / check_in / transfer / disposal
from_user_id	UUID	FK → users.id, NULLABLE	User returning or transferring the asset (check-in / transfer from)
to_user_id	UUID	FK → users.id, NULLABLE	User receiving the asset (check-out / transfer to)
from_location_id	UUID	FK → locations.id, NULLABLE	Location transferring from
to_location_id	UUID	FK → locations.id, NULLABLE	Location transferring to
from_department_id	UUID	FK → departments.id, NULLABLE	Department transferring from
to_department_id	UUID	FK → departments.id, NULLABLE	Department transferring to
transaction_date	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	When the transaction occurred
expected_return_date	DATE	NULLABLE	For check-outs: when the asset should be returned
actual_return_date	DATE	NULLABLE	For check-ins: when the asset was actually returned
authorising_user_id	UUID	FK → users.id, NOT NULL	Manager or officer who approved this transaction
purpose	TEXT	NULLABLE	Business purpose or justification for the transaction
condition_on_out	VARCHAR(30)	NULLABLE	Enum: good / damaged / incomplete (at time of check-out)
condition_on_in	VARCHAR(30)	NULLABLE	Enum: good / damaged / incomplete / faulty (at time of check-in)
incident_ticket_ref	VARCHAR(100)	NULLABLE	Reference to helpdesk ticket if damage reported on return
notes	TEXT	NULLABLE	Additional transaction notes
status	VARCHAR(20)	NOT NULL DEFAULT active	Enum: active / completed / overdue / cancelled
overdue_notified_at	TIMESTAMPTZ	NULLABLE	Timestamp when overdue notification was last sent

4.12 certificates Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Certificate asset identifier
asset_id	UUID	FK → assets.id, NOT NULL	Parent asset record in the core assets table
certificate_type	VARCHAR(50)	NOT NULL	Enum: ca_root / ca_intermediate / self_signed / wildcard / san / server / client / code_signing
issuing_authority	VARCHAR(255)	NOT NULL	Name of the Certificate Authority
subject_dn	TEXT	NOT NULL	Full X.509 Distinguished Name string
common_name	VARCHAR(255)	NOT NULL	CN component of the subject DN
san_entries	TEXT[]	NULLABLE	Subject Alternative Names (array)
serial_number	VARCHAR(100)	NOT NULL	Certificate serial number (hex)
fingerprint_sha256	CHAR(64)	NOT NULL	SHA-256 fingerprint for uniqueness and tamper detection
algorithm	VARCHAR(30)	NOT NULL	Enum: RSA-2048 / RSA-4096 / ECDSA-256 / ECDSA-384
issue_date	DATE	NOT NULL	Date certificate was issued
expiry_date	DATE	NOT NULL	Date certificate expires
days_remaining	INTEGER	COMPUTED DAILY	Calculated: expiry_date - CURRENT_DATE
status	VARCHAR(20)	NOT NULL	Enum: valid / review_soon / warning / critical / expired (auto-updated daily)
renewal_lead_days	INTEGER	DEFAULT 90	Days before expiry to trigger the renewal workflow
auto_renewal_enabled	BOOLEAN	DEFAULT FALSE	Whether automated renewal via ACME is enabled
environment	VARCHAR(20)	NOT NULL DEFAULT production	Enum: production / dr / staging / lab / development
key_store_location	TEXT	NULLABLE	Reference to HSM, keystore, or certificate vault
owner_user_id	UUID	FK → users.id, NULLABLE	Responsible person for renewal
previous_cert_id	UUID	FK → certificates.id, NULLABLE	Previous version of this certificate (renewal chain)
renewed_by_user_id	UUID	FK → users.id, NULLABLE	User who completed the most recent renewal
renewed_at	TIMESTAMPTZ	NULLABLE	Timestamp of the most recent renewal

4.13 licences Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Licence record identifier
asset_id	UUID	FK → assets.id, NOT NULL	Parent asset record in the core assets table
licence_type	VARCHAR(30)	NOT NULL	Enum: perpetual / subscription / volume / per_seat / per_core / trial / oem / open_source
total_seats	INTEGER	NULLABLE	Total purchased seats / licences
used_seats	INTEGER	DEFAULT 0	Currently assigned / in-use seats
available_seats	INTEGER	COMPUTED	Calculated: total_seats - used_seats
utilisation_pct	NUMERIC(5,2)	COMPUTED	Calculated: (used_seats / total_seats) * 100
expiry_date	DATE	NULLABLE	Mandatory for subscription and trial licences
days_remaining	INTEGER	COMPUTED DAILY	Calculated: expiry_date - CURRENT_DATE
status	VARCHAR(20)	NOT NULL	Enum: valid / review_soon / warning / critical / expired
renewal_lead_days	INTEGER	DEFAULT 90	Days before expiry to initiate renewal workflow
software_assurance_expiry	DATE	NULLABLE	Separate expiry date for Software Assurance or upgrade rights
key_or_reference	TEXT	NULLABLE	Encrypted licence key or vendor reference number
seat_alert_threshold_pct	SMALLINT	DEFAULT 85	Alert when seat utilisation exceeds this percentage

4.14 access_grants Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Access grant identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
access_type	VARCHAR(50)	NOT NULL	Enum: vendor_session / emergency_access / audit_access / elevated_rights / read_only_temp
target_asset_id	UUID	FK → assets.id, NULLABLE	Specific asset the access applies to (NULL = system-wide)

Column Name	Data Type	Constraints	Description
requestor_user_id	UUID	FK → users.id, NOT NULL	User requesting the access
approving_user_id	UUID	FK → users.id, NOT NULL	User who approved the access (must differ from requestor)
business_justification	TEXT	NOT NULL	Mandatory justification for the access grant
ticket_reference	VARCHAR(100)	NULLABLE	Helpdesk or change ticket reference
risk_level	VARCHAR(20)	NOT NULL DEFAULT medium	Enum: low / medium / high / critical
granted_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	When access was formally granted
expires_at	TIMESTAMPTZ	NOT NULL	Mandatory expiry timestamp
revoked_at	TIMESTAMPTZ	NULLABLE	When access was formally revoked (NULL = still active)
revoking_user_id	UUID	FK → users.id, NULLABLE	User who performed the revocation
revocation_confirmation	TEXT	NULLABLE	Confirmation note that access has been removed from target system
status	VARCHAR(20)	NOT NULL DEFAULT active	Enum: active / expired / revoked / overdue
overdue_escalated_at	TIMESTAMPTZ	NULLABLE	When overdue escalation notification was sent

4.15 change_requests Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Change request identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
change_reference	VARCHAR(50)	UNIQUE, NOT NULL	Human-readable change number (e.g., CHG-2026-0142)
target_asset_id	UUID	FK → assets.id, NOT NULL	Asset on which the change is being applied
change_type	VARCHAR(50)	NOT NULL	Enum: configuration / firmware_update / access_change / temporary_rule / other
description	TEXT	NOT NULL	Detailed description of the change being made
business_justification	TEXT	NOT NULL	Business reason for the change

Column Name	Data Type	Constraints	Description
risk_assessment	TEXT	NULLABLE	Risk analysis and potential impact description
revert_plan	TEXT	NOT NULL	Step-by-step instructions to undo this change
proposed_by_user_id	UUID	FK → users.id, NOT NULL	User proposing the change
approved_by_user_id	UUID	FK → users.id, NULLABLE	User who approved (must differ from proposer)
implemented_by_user_id	UUID	FK → users.id, NULLABLE	User who applied the change
verified_by_user_id	UUID	FK → users.id, NULLABLE	User who verified implementation
scheduled_start	TIMESTAMPTZ	NULLABLE	Planned implementation date/time
scheduled_revert_date	DATE	NOT NULL	Mandatory scheduled revert date for temporary changes
actual_applied_at	TIMESTAMPTZ	NULLABLE	When the change was actually implemented
actual_reverted_at	TIMESTAMPTZ	NULLABLE	When the change was actually reverted
days_until_revert	INTEGER	COMPUTED DAILY	Calculated: scheduled_revert_date - CURRENT_DATE
status	VARCHAR(30)	NOT NULL DEFAULT proposed	Enum: proposed / approved / applied / pending_revert / reverted / made_permanent / rejected / cancelled
is_overdue	BOOLEAN	COMPUTED	TRUE when scheduled_revert_date has passed and status = pending_revert
overdue_escalated_at	TIMESTAMPTZ	NULLABLE	When overdue notification was sent
made_permanent_reason	TEXT	NULLABLE	Justification if change was made permanent instead of reverted

4.16 notifications & notification_rules Tables

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Rule identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
event_type	VARCHAR(100)	NOT NULL	Triggering event (e.g., certificate.expiry, asset.overdue_return, access.expired)

Column Name	Data Type	Constraints	Description
trigger_days_before	INTEGER	NULLABLE	For time-based events: days before expiry/due date to fire
recipient_type	VARCHAR(30)	NOT NULL	Enum: specific_user / role / asset_owner / line_manager / department_head
recipient_id	UUID	NULLABLE	Specific user or role ID if recipient_type is specific
channels	TEXT[]	NOT NULL	Array: email / in_app / sms
subject_template	TEXT	NOT NULL	Notification subject with placeholder variables
body_template	TEXT	NOT NULL	Notification body with placeholder variables
is_active	BOOLEAN	DEFAULT TRUE	Whether this rule is currently active
cooldown_hours	INTEGER	DEFAULT 24	Minimum hours between repeat notifications for same event

5. AI Analytics Database Schema

> *The AI module introduces a dedicated schema that stores model registrations, prediction outputs, anomaly events, and recommendation records. All AI tables reference the core assets, users, and audit_logs tables.*

5.1 ai_models Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Model registration identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Tenant that owns this model configuration
model_name	VARCHAR(150)	NOT NULL	Human-readable model name (e.g., 'Licence Expiry Risk Scorer')
model_type	VARCHAR(50)	NOT NULL	Enum: classification / regression / anomaly_detection / forecasting / nlp / recommendation
model_version	VARCHAR(30)	NOT NULL	Semantic version of the model artefact
provider	VARCHAR(50)	NOT NULL	Enum: anthropic_claude / openai / internal / huggingface / custom
endpoint_url	TEXT	NULLABLE	API endpoint if model is hosted externally
api_key_secret_ref	TEXT	NULLABLE	Reference to secrets manager entry (never stored in plaintext)

Column Name	Data Type	Constraints	Description
input_features	JSONB	NOT NULL	Schema describing input fields the model expects
output_schema	JSONB	NOT NULL	Schema describing the model's output structure
training_dataset_ref	TEXT	NULLABLE	Reference to the training dataset used
accuracy_metric	NUMERIC(5,4)	NULLABLE	Latest measured accuracy or AUC score
threshold_config	JSONB	NULLABLE	Configurable thresholds for alerts and risk bands
retrain_schedule	VARCHAR(50)	NULLABLE	Cron expression for scheduled retraining
last_trained_at	TIMESTAMPTZ	NULLABLE	When the model was last trained or fine-tuned
last_run_at	TIMESTAMPTZ	NULLABLE	When the model last generated predictions
is_active	BOOLEAN	DEFAULT TRUE	Whether this model is currently running
is_approved	BOOLEAN	DEFAULT FALSE	Whether model has been approved for production use
approved_by_user_id	UUID	FK → users.id, NULLABLE	User who approved this model for production

5.2 ai_predictions Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Prediction record identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
model_id	UUID	FK → ai_models.id, NOT NULL	Model that generated this prediction
asset_id	UUID	FK → assets.id, NULLABLE	Asset the prediction relates to (NULL for org-wide predictions)
prediction_type	VARCHAR(100)	NOT NULL	Category: failure_risk / licence_overrun / expiry_forecast / lifecycle_recommendation / cost_anomaly
input_snapshot	JSONB	NOT NULL	Snapshot of input data used to generate this prediction
prediction_output	JSONB	NOT NULL	Raw output from the model
confidence_score	NUMERIC(5,4)	NULLABLE	Probability or confidence (0.0000–1.0000)

Column Name	Data Type	Constraints	Description
risk_band	VARCHAR(20)	NULLABLE	Enum: low / medium / high / critical (derived from score + thresholds)
predicted_value	TEXT	NULLABLE	Human-readable predicted value (e.g., failure within 30 days)
generated_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	When the prediction was produced
valid_until	TIMESTAMPTZ	NULLABLE	When this prediction is considered stale and should be refreshed
actioned_at	TIMESTAMPTZ	NULLABLE	When a user acknowledged or acted on this prediction
actioned_by_user_id	UUID	FK → users.id, NULLABLE	User who actioned the prediction
outcome	VARCHAR(30)	NULLABLE	Post-event outcome: correct / incorrect / inconclusive (for model feedback)
feedback_note	TEXT	NULLABLE	Free-text feedback from the actioning user

5.3 ai_anomalies Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Anomaly event identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
model_id	UUID	FK → ai_models.id, NOT NULL	Model that detected this anomaly
anomaly_type	VARCHAR(100)	NOT NULL	Category: usage_spike / bulk_export / after_hours_access / privilege_escalation / mass_deletion / cost_outlier
severity	VARCHAR(20)	NOT NULL	Enum: info / low / medium / high / critical
description	TEXT	NOT NULL	Plain-language description of the detected anomaly
affected_resource_type	VARCHAR(100)	NULLABLE	Type of resource (e.g., asset, user, report)
affected_resource_id	UUID	NULLABLE	UUID of the affected record
anomaly_score	NUMERIC(5,4)	NULLABLE	Anomaly score from the detection model
supporting_evidence	JSONB	NULLABLE	Data points that triggered the detection
detected_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	When the anomaly was detected

Column Name	Data Type	Constraints	Description
acknowledged_at	TIMESTAMPTZ	NULLABLE	When a user acknowledged the anomaly
acknowledged_by_user_id	UUID	FK → users.id, NULLABLE	User who acknowledged
resolution	TEXT	NULLABLE	Description of how the anomaly was resolved
resolved_at	TIMESTAMPTZ	NULLABLE	When the anomaly was formally resolved
false_positive	BOOLEAN	DEFAULT FALSE	Whether this was marked as a false positive
status	VARCHAR(20)	DEFAULT open	Enum: open / acknowledged / resolved / false_positive

5.4 ai_recommendations Table

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Recommendation identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
model_id	UUID	FK → ai_models.id, NOT NULL	Model that generated this recommendation
recommendation_type	VARCHAR(100)	NOT NULL	Category: decommission_asset / consolidate_licences / patch_now / renew_certificate / refresh_hardware / right_size_licence
priority	VARCHAR(20)	NOT NULL	Enum: low / medium / high / urgent
title	VARCHAR(255)	NOT NULL	Short recommendation title
description	TEXT	NOT NULL	Full explanation of the recommendation
estimated_saving	NUMERIC(15,2)	NULLABLE	Estimated cost saving if recommendation is implemented
estimated_risk_reduction	TEXT	NULLABLE	Description of risk reduction benefit
affected_asset_ids	UUID[]	NULLABLE	Assets the recommendation applies to
supporting_data	JSONB	NULLABLE	Data that supports the recommendation
generated_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	When the recommendation was produced
expires_at	TIMESTAMPTZ	NULLABLE	When this recommendation is no longer relevant

Column Name	Data Type	Constraints	Description
actioned_at	TIMESTAMPTZ	NULLABLE	When the recommendation was acted upon
actioned_by_user_id	UUID	FK → users.id, NULLABLE	User who acted on the recommendation
action_taken	TEXT	NULLABLE	Description of the action taken
status	VARCHAR(20)	DEFAULT active	Enum: active / actioned / dismissed / expired

5.5 ai_chat_sessions Table

Stores conversational AI interactions with users, enabling a natural-language interface for querying asset data, generating reports, and receiving recommendations.

Column Name	Data Type	Constraints	Description
id	UUID	PK, NOT NULL	Session identifier
organisation_id	UUID	FK → organisations.id, NOT NULL	Owning tenant
user_id	UUID	FK → users.id, NOT NULL	User who initiated the session
session_title	VARCHAR(255)	NULLABLE	Auto-generated or user-set title for the conversation
messages	JSONB	NOT NULL	Ordered array of {role, content, timestamp} message objects
context_asset_ids	UUID[]	NULLABLE	Asset IDs loaded into context for this session
model_used	VARCHAR(100)	NOT NULL	AI model identifier used for this session
total_tokens_used	INTEGER	DEFAULT 0	Cumulative token usage for cost tracking
started_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	Session start timestamp
last_activity_at	TIMESTAMPTZ	NOT NULL	Timestamp of last message in session
is_archived	BOOLEAN	DEFAULT FALSE	Whether the session has been archived by the user

6. Asset Category Field Schemas

Each asset category has a dedicated extended table (asset_extended_{category}) that stores category-specific fields, joined to the core assets table via asset_id. The M/O column indicates Mandatory (M) or Optional (O). The DB Column and Data Type columns show the physical schema.

6.1 Software Applications — asset_extended_applications

Field Name	M/O	Description & Validation	DB Column	Data Type
Application Name	M	Full commercial name; unique within the register per location	app_name	VARCHAR(255)
Version Number	M	Semantic version string (e.g., 3.2.1); updated on every upgrade	version_number	VARCHAR(50)
Version Release Date	M	Date the installed version was released by the vendor	version_release_date	DATE
Hosting Location Type	M	Enum: on_premises / cloud / hybrid / saas	hosting_type	VARCHAR(20)
Hosting Institution	M	Entity hosting the application if externally hosted	hosting_institution	VARCHAR(255)
Function / Purpose	M	Business function the application serves	function_description	TEXT
Licence Reference	M	FK to licences.id	licence_id	UUID
Number of Licences	M	Total licences purchased; must match the licence record	licence_count	INTEGER
Used Seats	M	Currently assigned seats; updated via Check-In/Out module	used_seats	INTEGER
Licence Type	M	Enum: perpetual / subscription / volume / per_seat / per_core / trial / oem / open_source	licence_type	VARCHAR(30)
Licence Expiry Date	M*	Mandatory for subscription/trial. Alerts at 180, 90, 30d	licence_expiry_date	DATE
Support Contract Ref	O	Vendor support or maintenance agreement reference	support_contract_ref	VARCHAR(100)
Support Expiry Date	O	Date vendor support ends; may differ from licence expiry	support_expiry_date	DATE
Related Server Assets	O	FK array to assets.id for servers hosting this application	host_server_ids	UUID[]
Related Database Assets	O	FK array to assets.id for databases used by this application	database_ids	UUID[]

6.2 Databases — asset_extended_databases

Field Name	M/O	Description & Validation	DB Column	Data Type
Database Engine	M	Enum: mssql / oracle / postgresql / mysql / mariadb / mongodb / other	db_engine	VARCHAR(30)
Engine Version	M	Version of the DBMS engine (e.g., PostgreSQL 16.2)	engine_version	VARCHAR(50)
Patch Level	O	Current cumulative patch or service pack level	patch_level	VARCHAR(50)
Host Server	M	FK to assets.id of the server hosting this database	host_server_id	UUID
Parent Application(s)	M	FK array to assets.id for applications using this database	parent_application_ids	UUID[]
Data Classification	M	Enum: top_secret / secret / confidential / internal / public	data_classification	VARCHAR(20)
Backup Schedule	M	Frequency and type (e.g., full_daily + incremental_hourly)	backup_schedule	TEXT
Last Backup Verified	M	Date a successful restore test was last completed	last_backup_verified_date	DATE
Encryption at Rest	M	Enum: yes / no / partial	encryption_at_rest	VARCHAR(10)
Encryption in Transit	M	Boolean; TLS version if applicable	encryption_in_transit	BOOLEAN
TLS Version	O	TLS version in use for in-transit encryption	tls_version	VARCHAR(10)
RTO (minutes)	O	Recovery Time Objective in minutes from Business Continuity Plan	rto_minutes	INTEGER
RPO (minutes)	O	Recovery Point Objective in minutes	rpo_minutes	INTEGER
Licence Reference	O	FK to licences.id for the DBMS licence	licence_id	UUID

6.3 Servers & Virtualisation — asset_extended_servers

Field Name	M/O	Description & Validation	DB Column	Data Type
Server Type	M	Enum: physical / virtual_machine / cloud_instance / container_host / hypervisor	server_type	VARCHAR(30)
Hypervisor Platform	O	Enum: vmware / hyper_v / proxmox / nutanix / kvm / other	hypervisor_platform	VARCHAR(30)

Field Name	M/O	Description & Validation	DB Column	Data Type
VM Host Server	O	FK to assets.id of the physical host (for VMs)	host_server_id	UUID
Make / Manufacturer	M	Hardware vendor (Dell, HPE, Cisco UCS, etc.)	manufacturer	VARCHAR(100)
Model Number	M	Exact hardware model (e.g., Dell PowerEdge R750)	model_number	VARCHAR(150)
Serial Number	M	Manufacturer serial number; for physical assets	serial_number	VARCHAR(100)
Operating System	M	Full OS name and edition	os_name	VARCHAR(150)
OS Version / Build	M	Specific build or kernel version	os_version	VARCHAR(100)
OS Release Date	M	Date the installed OS version was released	os_release_date	DATE
OS End-of-Support Date	O	Vendor mainstream support end date; alerts at 180d	os_eos_date	DATE
Last OS Patch Date	M	Date the most recent security patch was applied	last_patch_date	DATE
Next Patch Window	O	Scheduled next maintenance / patching date	next_patch_date	DATE
CPU Count	M	Total physical or virtual CPU sockets	cpu_count	SMALLINT
Total Cores	M	Total CPU cores allocated	total_cores	SMALLINT
RAM (GB)	M	Total RAM installed or allocated	ram_gb	INTEGER
Storage (GB)	M	Total primary storage allocated	storage_gb	INTEGER
Storage Type	O	Enum: ssd / hdd / san / nas / nvme	storage_type	VARCHAR(20)
Management IP Address	M	Primary management IP in CIDR notation	mgmt_ip_address	INET
VLAN Assignment	O	VLAN(s) the server is connected to	vlan_assignment	VARCHAR(255)
Server Role	M	Enum: domain_controller / database / application / file / monitoring / other	server_role	VARCHAR(50)
DR Site Counterpart	O	FK to assets.id of the DR replica	dr_counterpart_id	UUID
Backup Policy Reference	M	Reference to the backup policy applied to this server	backup_policy_ref	VARCHAR(100)

6.4 Network Devices — asset_extended_network_devices

Field Name	M/O	Description & Validation	DB Column	Data Type
Device Category	M	Enum: access_switch / distribution_switch / core_switch / router / firewall / wireless_ap / wireless_controller / load_balancer / ids_ips / vpn_concentrator / other	device_category	VARCHAR(50)
Manufacturer	M	Cisco / Juniper / Aruba / Fortinet / Palo Alto / Extreme / Other	manufacturer	VARCHAR(100)
Model Number	M	Exact hardware model	model_number	VARCHAR(150)
Serial Number	M	Hardware serial number	serial_number	VARCHAR(100)
Management IP Address	M	Primary management IP in IPv4 or IPv6 CIDR	mgmt_ip_address	INET
IOS / Firmware Version	M	Current operating firmware version	firmware_version	VARCHAR(100)
Firmware Release Date	M	Date the installed firmware was released	firmware_release_date	DATE
Last Firmware Update	M	Date firmware was last patched or upgraded	last_firmware_update	DATE
Firmware End-of-Support	O	Date vendor ends support for the installed firmware; alert at 180d	firmware_eos_date	DATE
Next Firmware Window	O	Scheduled next patching window	next_firmware_window	DATE
ISP / Carrier	O	For WAN-facing devices: ISP or carrier circuit provider	isp_carrier	VARCHAR(150)
Licence Type	M	Enum: perpetual_hardware / smartnet / dna_subscription / security_licence / evaluation	licence_type	VARCHAR(50)
Licence Reference	O	FK to licences.id for device software licence	licence_id	UUID
Licence Expiry Date	O	Expiry of device-specific subscription licence	licence_expiry_date	DATE
802.1X / NAC Status	O	Enum: configured / partial / not_configured	nac_status	VARCHAR(20)
VLAN Assignment	O	VLANs the device participates in (comma-separated)	vlan_assignment	TEXT
Uplink Device(s)	O	FK array to upstream device asset records	uplink_device_ids	UUID[]
Downlink Device(s)	O	FK array to downstream device or server records	downlink_device_ids	UUID[]
Physical Rack Location	O	FK to locations.id at rack level	rack_location_id	UUID

Field Name	M/O	Description & Validation	DB Column	Data Type
Support Contract Ref	O	Vendor support or SmartNet contract reference	support_contract_ref	VARCHAR(100)

6.5 Client Devices — asset_extended_client_devices

Field Name	M/O	Description & Validation	DB Column	Data Type
Device Type	M	Enum: desktop / laptop / tablet / ip_phone / printer / scanner / mobile / pos_terminal / other	device_type	VARCHAR(30)
Manufacturer	M	Dell / HP / Lenovo / Apple / Cisco / Other	manufacturer	VARCHAR(100)
Model	M	Full model name and number	model	VARCHAR(150)
Serial Number	M	Hardware serial number	serial_number	VARCHAR(100)
MAC Address (Wired)	M	Wired network interface MAC address	mac_wired	MACADDR
MAC Address (Wireless)	O	Wireless interface MAC address	mac_wireless	MACADDR
Year of Purchase	M	Year the device was procured	year_of_purchase	SMALLINT
Year of Deployment	M	Year the device was placed into active service	year_of_deployment	SMALLINT
Operating System	M	Full OS name and version installed	os_name	VARCHAR(150)
OS Patch Level	M	Most recent OS patch or update applied	os_patch_level	VARCHAR(100)
Network VLAN	O	VLAN the device is connected to	vlan	SMALLINT
Switch Port	O	Switch hostname and port (e.g., SW01 Gi1/0/5)	switch_port	VARCHAR(100)
Disk Encryption	M	Boolean; full-disk encryption status	disk_encrypted	BOOLEAN
Encryption Tool	O	BitLocker / FileVault / VeraCrypt / Other	encryption_tool	VARCHAR(50)
Antivirus / EDR	M	Name and version of endpoint protection installed	av_edr_name	VARCHAR(150)
Last Vulnerability Scan	M	Date of the most recent vulnerability scan	last_vuln_scan_date	DATE
Vulnerability Scan Source	O	Tool used for scan (Tenable, Qualys, Defender, etc.)	vuln_scan_source	VARCHAR(100)

Field Name	M/O	Description & Validation	DB Column	Data Type
Check-In/Out History	**M**	Auto-populated from check_in_out_transactions; read-only in this view	[computed]	[join]

7. Functional Requirements

> *Requirements are classified as: Mandatory (system cannot be accepted without it), High (required in first release), Medium (within six months of go-live), or Low (future release desirable).*

7.1 Asset Registry Module (FR-AR)

Req. ID	Priority	Requirement
FR-AR-001	Mandatory	The system shall maintain a unified, searchable registry of all IT assets across all configured asset categories. The number and type of categories shall be configurable per organisation.
FR-AR-002	Mandatory	Each asset record shall be assigned a globally unique system-generated Asset ID at creation. The ID format shall be configurable per asset category using a template (e.g., {CODE}-{LOC}-{YYYY}-{SEQ:4}).
FR-AR-003	Mandatory	The system shall enforce all mandatory fields defined in the asset schema for each category before a record can be saved. Mandatory fields are configurable per asset category.
FR-AR-004	Mandatory	The system shall support import of existing asset records via CSV and Excel upload, with field mapping, validation error reporting, and duplicate detection before data is committed.
FR-AR-005	Mandatory	Assets shall be linkable to form a Configuration Management Database (CMDB). Supported relationship types shall include: runs_on, depends_on, is_component_of, is_backed_up_by, connects_to, hosts, and secures.
FR-AR-006	Mandatory	Every field change to any asset record shall be captured in the immutable audit log (Section 4.10), recording: field name, previous value, new value, actor identity, timestamp, and IP address.
FR-AR-007	Mandatory	The system shall support an unlimited number of custom fields per asset category, configurable by an administrator without code changes (see asset_categories and custom_fields tables in Section 4).
FR-AR-008	High	The system shall support asset lifecycle status transitions. Backward transitions (e.g., Decommissioned to Active) shall require an authorised override and shall be logged in the audit trail.
FR-AR-009	High	Assets shall be organisable using a configurable location hierarchy (Organisation → Region → Branch → Building → Floor → Room → Rack). The hierarchy levels and names shall be configurable per organisation.
FR-AR-010	High	The system shall support bulk update operations with a mandatory approval workflow and full audit logging of all records changed in each operation.

Req. ID	Priority	Requirement
FR-AR-011	High	The system shall support attachment of supporting documents (purchase orders, warranties, configuration exports, photographs) to any asset record, with a configurable maximum file size.
FR-AR-012	Medium	The system shall provide a network topology visualisation showing physical and logical relationships between network devices, servers, and hosted applications.
FR-AR-013	Medium	The system shall support RFID and barcode / QR code scanning for physical asset discovery, registration, and check-in/out via mobile browser or dedicated mobile application.
FR-AR-014	Low	The system shall support automatic asset discovery via SNMP v3 and WMI agents for network-connected devices, flagging unregistered assets for review.

7.2 Asset Check-In / Check-Out Module (FR-CO)

Req. ID	Priority	Requirement
FR-CO-001	Mandatory	The system shall support formal check-out of any asset to a named user, department, location, or project. Check-out shall require electronic approval from an authorising officer.
FR-CO-002	Mandatory	The system shall record all transaction fields defined in the check_in_out_transactions table (Section 4.11) for every check-out, check-in, and transfer event.
FR-CO-003	Mandatory	The system shall send automated notifications to the assigned user and their line manager when an asset approaches its expected return date at configurable intervals (default: 7 days and 1 day prior).
FR-CO-004	Mandatory	The system shall flag assets checked out beyond their authorised period as Overdue and escalate to the Asset Manager and line manager via automated notification.
FR-CO-005	High	The system shall maintain a complete chronological assignment history for every asset from registration to the present date, accessible via the asset record.
FR-CO-006	High	The system shall support inter-location asset transfers requiring confirmation from both the sending and receiving location officer, creating a two-step acknowledgement trail.
FR-CO-007	High	Users shall be able to request assets through a self-service portal, triggering a configurable approval workflow before check-out is permitted.
FR-CO-008	Medium	The system shall support consumable asset tracking with configurable quantities, reorder thresholds, and consumption history.

7.3 Certificate and Licence Management Module (FR-CL)

Req. ID	Priority	Requirement
FR-CL-001	Mandatory	The system shall maintain a dedicated Certificate Register linked to the core asset registry. Every certificate record shall be associated with its dependent asset records via asset_relationships.
FR-CL-002	Mandatory	The system shall calculate days_remaining for every certificate and licence daily and apply status bands: EXPIRED / CRITICAL (≤30d) / WARNING (≤90d) / REVIEW SOON (≤180d) / VALID.
FR-CL-003	Mandatory	The system shall generate automated multi-channel alerts at configurable intervals before certificate or licence expiry (defaults: 180, 90, 30, 14, 7, 1 day). Alert recipients shall be configurable per-certificate.
FR-CL-004	Mandatory	The system shall initiate a renewal workflow automatically when a certificate or licence reaches its configured renewal lead time, assigning the task to the nominated owner with a target completion date.
FR-CL-005	Mandatory	Licence seat utilisation shall be tracked in real time. The system shall alert when utilisation exceeds a configurable threshold (default: 85% of total seat count).
FR-CL-006	High	The system shall display a Renewal Calendar view showing all certificates and licences expiring within the next configurable window (default: 12 months), sorted by urgency, with export to PDF.
FR-CL-007	High	The system shall maintain a full version history for every certificate, capturing serial number, fingerprint, issuer, and the officer who completed each renewal.
FR-CL-008	High	The system shall support integration with ACME protocol endpoints and Windows ADCS for automated certificate discovery, status polling, and renewal initiation.
FR-CL-009	Medium	The system shall track perpetual licences separately from subscription licences and shall separately alert on the expiry of Software Assurance and support contract periods.

7.4 Temporary Access and Configuration Change Management (FR-AC)

Req. ID	Priority	Requirement
FR-AC-001	Mandatory	The system shall provide a Temporary Access Log for recording all temporary privileged access grants, using the field schema defined in the access_grants table (Section 4.14).
FR-AC-002	Mandatory	The system shall automatically flag access grants that have passed their expiry timestamp and have not been formally revoked, escalating to the Security Officer within a configurable period (default: 1 hour).
FR-AC-003	Mandatory	Access revocation shall require a formal entry recording the revoking officer, date/time, and confirmation of access removal. The system shall enforce that a user cannot revoke their own access grant.
FR-AC-004	Mandatory	The system shall maintain a Configuration Change Log using the schema defined in the change_requests table (Section 4.15). Every change entry shall include a mandatory Scheduled Revert Date.
FR-AC-005	Mandatory	The system shall calculate days_until_revert for every pending change and shall flag overdue reverts in a prominent OVERDUE state with automated escalation.

Req. ID	Priority	Requirement
FR-AC-006	High	Configuration changes shall progress through a configurable approval workflow: Proposed → Approved → Applied → Pending Revert → Reverted or Made Permanent. Each transition shall require a named authoriser.
FR-AC-007	High	The system shall enforce at the workflow engine level that the same user cannot both propose and approve a configuration change, regardless of their role assignments.
FR-AC-008	Medium	The system shall support integration with network management platforms (e.g., Cisco DNA Center, SolarWinds) to automatically import change events and match them against open change log entries.

7.5 User and People Management Module (FR-UM)

Req. ID	Priority	Requirement
FR-UM-001	Mandatory	The system shall maintain a User Register using the schema defined in the users table (Section 4.4), linking all users to their assigned assets, system roles, and organisational structure.
FR-UM-002	Mandatory	When a user's employment status changes to Terminated or Suspended, the system shall automatically trigger a configurable off-boarding checklist requiring return of all assigned assets and access revocation within a configurable period (default: 24 hours).
FR-UM-003	Mandatory	The system shall integrate with Active Directory / LDAP for user authentication and record synchronisation. User records shall not require manual duplication.
FR-UM-004	Mandatory	The system shall support SAML 2.0 and OpenID Connect for Single Sign-On with any compatible identity provider (Azure AD, Okta, on-premises ADFS, Keycloak, Google Workspace, or any SAML/OIDC provider).
FR-UM-005	High	The system shall provide a 'view as user' report showing every asset and access right attributed to a specific individual, for instant access review during audits.
FR-UM-006	High	Role assignments shall require approval from a second administrator. No user shall be able to assign their own roles or escalate their own privileges.
FR-UM-007	High	The system shall support contractor and vendor user records with mandatory expiry dates, automatic access revocation on expiry, and a configurable re-approval cycle.
FR-UM-008	Medium	The system shall support a configurable organisational hierarchy for line management reporting, enabling automated escalation routing without manual configuration of each escalation path.

7.6 AI-Powered Analytics and Intelligence Module (FR-AI)

> *This module is the primary enhancement over prior versions of this specification. All AI model configurations, thresholds, and prompt templates are adjustable in the Administration Console without code changes.*

Req. ID	Priority	Requirement
FR-AI-001	Mandatory	The system shall embed an AI analytics engine supporting the following analytical capabilities: predictive failure risk scoring, licence overrun forecasting, anomaly detection in audit logs, lifecycle optimisation recommendations, and natural-language asset querying.
FR-AI-002	Mandatory	The system shall provide an AI-powered Executive Dashboard summarising key insights in plain language: assets at risk of failure, certificates expiring within 30 days, unusual access patterns, and cost optimisation opportunities.
FR-AI-003	Mandatory	The system shall maintain a Predictions Register (ai_predictions table) where every AI-generated prediction is stored, traceable to the model that produced it, and reviewable by authorised users.
FR-AI-004	Mandatory	The AI module shall support a natural-language chat interface allowing authorised users to query the asset register in plain language (e.g., 'Which servers have not been patched in the last 90 days?' or 'Show me all critical assets in the Nairobi branch') without requiring knowledge of the underlying schema.
FR-AI-005	Mandatory	All AI-generated insights, predictions, and anomaly alerts shall be logged in the audit trail (audit_logs table) to maintain full traceability of AI-assisted decisions.
FR-AI-006	High	The system shall perform predictive asset lifecycle analysis, estimating the probability of hardware failure or end-of-life status for each physical asset based on age, warranty status, historical incident records, and vendor end-of-support dates.
FR-AI-007	High	The system shall perform licence utilisation forecasting, predicting the date at which current licence seats will be exhausted based on historical usage growth trends, enabling proactive procurement before a breach occurs.
FR-AI-008	High	The system shall perform anomaly detection on the audit log stream, flagging unusual patterns including: bulk data exports outside business hours, repeated failed login attempts, mass record deletion events, and privilege escalation sequences.
FR-AI-009	High	The system shall generate cost optimisation recommendations, identifying: underutilised licences eligible for downgrade, assets approaching end-of-life that should be budgeted for replacement, and duplicate software licences that could be consolidated.
FR-AI-010	High	The system shall support an AI model registry (ai_models table) allowing administrators to register, configure, version, and approve AI models for production use. Multiple models may be registered, and each can be assigned to specific analytical tasks.
FR-AI-011	High	The system shall provide a model performance dashboard showing: prediction accuracy over time, anomaly detection precision and recall, false positive rates, and model drift indicators, enabling data tE-AMS to monitor and retrain models as required.
FR-AI-012	High	AI-generated recommendations shall be presented with: a priority level, a plain-language explanation of supporting evidence, an estimated cost saving or risk reduction value, and a one-click action pathway to initiate the recommended action.
FR-AI-013	Medium	The system shall support automated report narration: when generating any standard report, an AI model shall produce a plain-English executive summary highlighting the most significant findings, trends, and required actions.
FR-AI-014	Medium	The system shall support configurable AI alerting thresholds and confidence score cutoffs per model, allowing organisations to tune sensitivity without retraining the model.
FR-AI-015	Medium	The system shall implement a human-in-the-loop feedback mechanism: users who action, dismiss, or mark a prediction as a false positive shall be able to provide structured

Req. ID	Priority	Requirement
		feedback that is stored in the ai_predictions and ai_anomalies tables and used to improve model accuracy over time.
FR-AI-016	Low	The system shall support predictive budget forecasting, using historical asset acquisition patterns and upcoming warranty / licence expiry data to generate a projected IT budget requirement for the next 12–36 months.
FR-AI-017	Low	The system shall support integration with external AI/ML platforms (e.g., AWS SageMaker, Azure ML, Google Vertex AI) via the ai_models table endpoint configuration, allowing organisations to host and version models in their preferred ML infrastructure.

7.7 Reporting and Analytics Module (FR-RP)

Req. ID	Priority	Requirement
FR-RP-001	Mandatory	The system shall provide a real-time executive dashboard displaying: total asset counts by category, certificate/licence expiry status distribution, open change requests, overdue reverts, and pending access revocations.
FR-RP-002	Mandatory	The system shall generate a complete Asset Inventory Report exportable to PDF, Excel (XLSX), and CSV, filterable by any combination of asset category, location, department, status, and date range.
FR-RP-003	Mandatory	The system shall generate a Software Licence Compliance Report showing: total seats purchased, seats in use, available seats, utilisation percentage, and expiry date for every licence.
FR-RP-004	Mandatory	The system shall generate an IT Audit Report for any selected time period, including all asset changes, access grants and revocations, configuration changes and their revert status, and certificate renewals.
FR-RP-005	Mandatory	All reports shall carry a digital timestamp, the name of the generating user, the data verification date, and a configurable watermark, enabling use as audit evidence documents.
FR-RP-006	High	The system shall provide a multi-location comparison dashboard showing asset health, compliance posture, and outstanding actions across all locations simultaneously.
FR-RP-007	High	The system shall support scheduled report delivery, automatically emailing specified reports to configured recipients on a defined schedule (daily / weekly / monthly).
FR-RP-008	High	The system shall provide asset depreciation tracking and generate a Depreciation Report using configurable useful life periods and acquisition costs per asset category.
FR-RP-009	High	The system shall provide an AI-enhanced Security Posture Report covering: certificate health, licence compliance rate, open temporary access grants, overdue configuration reverts, anomaly detection findings, and predictive risk scores.
FR-RP-010	Medium	The system shall support a custom report builder allowing authorised users to define report templates selecting any combination of fields, filters, groupings, and date ranges without technical intervention.

7.8 Audit Log and Compliance Module (FR-AU)

Req. ID	Priority	Requirement
FR-AU-001	Mandatory	The system shall maintain an immutable, append-only audit log using the schema defined in the audit_logs table (Section 4.10). No UPDATE or DELETE operation shall be permitted on this table by any system user including administrators.
FR-AU-002	Mandatory	Each audit log entry shall be cryptographically signed (HMAC-SHA256) using a rotating secret key to enable tampering detection.
FR-AU-003	Mandatory	The audit log shall retain entries for a configurable minimum period (default: 7 years) with automated secure archival after the first year.
FR-AU-004	Mandatory	The system shall support a formal Annual Audit Review workflow in which an auditor records findings, actions taken, reviewer identity, and next review date, creating a persistent compliance review record.
FR-AU-005	High	The audit log shall be searchable and filterable by date range, user, action type, asset category, and specific record. Search results shall be exportable as evidence packages in PDF format.
FR-AU-006	High	The system shall generate a periodic Audit Compliance Certificate per asset category showing the percentage of records with a last audit date within the required frequency (configurable by criticality band).
FR-AU-007	High	The AI anomaly detection engine (FR-AI-008) shall monitor the audit log stream in near-real-time and surface alerts within the Audit Module dashboard for all detected anomalous patterns.
FR-AU-008	Medium	The system shall integrate with SIEM platforms (Splunk, IBM QRadar, Microsoft Sentinel, or any syslog-TLS receiver) by forwarding all audit log entries in CEF or JSON format.

7.9 Notification and Workflow Engine (FR-NW)

Req. ID	Priority	Requirement
FR-NW-001	Mandatory	The system shall support configurable notification rules using the schema defined in the notification_rules table (Section 4.16) for all time-based and event-based triggers.
FR-NW-002	Mandatory	Notifications shall be deliverable via in-application alert, email (SMTP), and SMS gateway. Each channel shall be independently configurable per event type and per recipient.
FR-NW-003	Mandatory	Multi-step approval workflows shall be supported for: asset check-out requests, access grant requests, configuration change proposals, role assignments, and asset disposal approvals.
FR-NW-004	High	Workflow steps shall enforce segregation of duties rules at the engine level, preventing a configured approver from also being the initiator of the same workflow instance.
FR-NW-005	High	Workflow escalation shall be automatic: if a pending approval exceeds the configured response deadline, the request shall be escalated to the approver’s line manager with a copy to the Security Officer.

Req. ID	Priority	Requirement
FR-NW-006	High	The system shall provide a personal 'My Actions' queue consolidating all pending approvals, outstanding tasks, and overdue items for each user.
FR-NW-007	Medium	The system shall support webhook-based notifications, allowing integration with external collaboration platforms (Microsoft TE-AMS, Slack, or any webhook endpoint) for event-driven alerts.

7.10 System Configuration and Customisation Module (FR-SY)

Req. ID	Priority	Requirement
FR-SY-001	Mandatory	The system shall provide an Administration Console through which all configurable system parameters may be adjusted by authorised administrators without code changes or system restarts.
FR-SY-002	Mandatory	The Administration Console shall allow creation, modification, and deactivation of asset categories, field schemas, mandatory/optional field designations, and Asset ID format templates.
FR-SY-003	Mandatory	The Administration Console shall support full white-label branding configuration: organisation name, logo, colour scheme, and custom domain, applied at the tenant level.
FR-SY-004	Mandatory	The system shall support multi-tenancy, with complete data isolation between tenants enforced at the database level via organisation_id scoping on all tables.
FR-SY-005	High	The Administration Console shall allow configuration of RBAC roles, permissions, and permission scopes (per asset category or per location) without redeployment.
FR-SY-006	High	The Administration Console shall provide a Workflow Designer enabling administrators to define custom approval workflows, step sequences, and escalation paths for any configurable event type.
FR-SY-007	High	The system shall support configuration of the organisational hierarchy levels, names, and codes (e.g., an organisation may label their levels as 'Division', 'District', 'Site' rather than the defaults).
FR-SY-008	High	All configuration changes made in the Administration Console shall be logged in the audit trail with full before/after values.
FR-SY-009	Medium	The system shall support a configuration export/import function, allowing organisations to export their full configuration as a versioned JSON file and import it into a new or restored instance.
FR-SY-010	Medium	The system shall support multiple currencies and automatic currency conversion for financial fields using configurable exchange rates updated at a configurable frequency.

8. Non-Functional Requirements

8.1 Security Requirements (NFR-SEC)

Req. ID	Priority	Requirement
NFR-SEC-001	Mandatory	The system shall enforce Multi-Factor Authentication for all user logins. Supported factors shall include: TOTP (RFC 6238), hardware FIDO2/WebAuthn tokens, and SMS OTP as a fallback. MFA requirements per role shall be configurable.
NFR-SEC-002	Mandatory	All data in transit shall be encrypted using TLS 1.2 minimum, with TLS 1.3 as the preferred standard. All HTTP traffic shall be redirected to HTTPS. Weak cipher suites shall be explicitly disabled.
NFR-SEC-003	Mandatory	All data at rest shall be encrypted using AES-256. Database fields classified as sensitive shall be additionally encrypted at the field level, with key management via a dedicated KMS or HSM.
NFR-SEC-004	Mandatory	The system shall implement RBAC with the principle of least privilege. Role and permission configurations shall be auditable at any point in time.
NFR-SEC-005	Mandatory	User sessions shall expire after a configurable inactivity period (default: 15 minutes for privileged roles, 30 minutes for standard roles). Session tokens shall be invalidated on logout.
NFR-SEC-006	Mandatory	The system shall enforce a configurable password policy (minimum length, complexity, history, maximum age) applied to all users not authenticated exclusively via SSO.
NFR-SEC-007	Mandatory	Account lockout shall activate after a configurable number of consecutive failed authentication attempts (default: 5). Locked accounts shall require manual administrator unlock and shall generate an immediate security alert.
NFR-SEC-008	Mandatory	The system shall implement OWASP-recommended security headers on all web responses, including CSP, HSTS, X-Frame-Options, X-Content-Type-Options, and Referrer-Policy.
NFR-SEC-009	High	All API endpoints shall be authenticated using OAuth 2.0 with configurable scopes, or API key authentication with key rotation support. API access shall be logged equivalently to UI access.
NFR-SEC-010	High	The system shall pass an independent OWASP Top 10 penetration test before go-live and annually thereafter. Critical and high vulnerabilities shall be remediated within 30 days.
NFR-SEC-011	High	AI model API keys and external service credentials shall be stored in an integrated secrets manager and shall never be stored in plaintext in the database or configuration files.

8.2 Performance and Scalability (NFR-PERF)

Req. ID	Priority	Requirement
NFR-PERF-001	Mandatory	The system shall support a minimum of 500 concurrent active user sessions without performance degradation, scaling to 5,000+ concurrent users via horizontal scaling.
NFR-PERF-002	Mandatory	Page load time for any standard dashboard or asset list view shall not exceed 3 seconds for 95% of requests under normal load, measured at the server.
NFR-PERF-003	Mandatory	The system shall support an asset register of at least 500,000 asset records without search or report generation exceeding 10 seconds.

Req. ID	Priority	Requirement
NFR-PERF-004	High	AI prediction batch jobs shall be scheduled during off-peak windows and shall not degrade the response time of interactive user sessions during business hours.
NFR-PERF-005	High	The system shall be architected for horizontal scaling by adding application nodes behind a load balancer, without architectural changes or extended downtime.

8.3 Availability and Business Continuity (NFR-AV)

Req. ID	Priority	Requirement
NFR-AV-001	Mandatory	The production instance shall achieve minimum availability of 99.5% per calendar month, excluding pre-announced maintenance windows.
NFR-AV-002	Mandatory	The system shall be deployed in an active-passive HA configuration with automatic failover to a DR instance within a configurable period (default RTO ≤ 15 minutes).
NFR-AV-003	Mandatory	Database backups shall be performed on a configurable schedule (default: full daily, transaction log every 15 minutes). Recovery Point Objective shall not exceed 15 minutes.
NFR-AV-004	Mandatory	A full restore test from backup shall be performed on a configurable schedule (default: quarterly) and formally documented in the audit log.
NFR-AV-005	High	Planned maintenance windows shall be schedulable by administrators. Users shall receive configurable advance notification (default: 72 hours) before any planned maintenance event.

8.4 Usability and Accessibility (NFR-UX)

Req. ID	Priority	Requirement
NFR-UX-001	Mandatory	The system shall provide a web-based UI accessible on current versions of Chrome, Firefox, Edge, and Safari without requiring plugins or browser extensions.
NFR-UX-002	Mandatory	The web interface shall be fully responsive and functional on tablet devices (minimum 768px viewport width).
NFR-UX-003	Mandatory	The system shall conform to WCAG 2.1 Level AA accessibility standards.
NFR-UX-004	High	The system shall support full localisation (i18n), initially in English, with an architecture permitting additional languages without code changes.
NFR-UX-005	High	The system shall provide contextual help text and field-level tooltips for all mandatory fields.
NFR-UX-006	Medium	A dedicated mobile application (iOS and Android) shall be provided for field operations: barcode/QR scanning, check-in/out, quick asset registration, and AI assistant access.

Req. ID	Priority	Requirement
NFR-UX-007	Medium	The AI chat interface shall be accessible from any screen within the system via a persistent assistant panel, with the ability to scope queries to the asset or report currently being viewed.

9. Integration Requirements

All integrations shall use authenticated, encrypted API communication with full logging of every integration event in the system audit trail. Integration credentials shall be stored in the secrets manager referenced in NFR-SEC-011.

Integration Target	Protocol / Standard	Purpose
Active Directory / LDAP	LDAP / LDAPS over TLS	User authentication, user record synchronisation, group membership for role mapping
SAML 2.0 / OIDC SSO	SAML 2.0 / OIDC	Single Sign-On for any compatible identity provider (Azure AD, Okta, Google, ADFS, Keycloak)
Network Access Control (NAC)	RADIUS / TACACS+ / REST	Verify 802.1X authentication events; import device authentication status
Network Management Platform	REST API	Import network device inventory, firmware versions, topology, and configuration compliance status (Cisco DNA Center, SolarWinds, Zabbix, PRTG)
Endpoint Management	REST API / WMI	Import software inventory, patch compliance, and device health (SCCM / Microsoft Intune / Jamf)
Vulnerability Scanner	REST API	Import scan results and link findings to specific asset records (Tenable Nessus, Qualys, Defender for Endpoint)
SIEM Platform	Syslog TLS / CEF / JSON	Forward all audit log entries for centralised security monitoring (Splunk, QRadar, Sentinel, Elastic SIEM)
Certificate Authority / ACME	REST API / ACME Protocol	Certificate discovery, automated renewal, and status polling
Financial / ERP System	REST API	Pull purchase order data, contract values, and acquisition costs for financial reporting
Email Server	SMTP with TLS / STARTTLS	Delivery of automated notifications, alerts, and approval requests
SMS Gateway	HTTPS REST	Delivery of MFA OTPs, critical expiry alerts, and emergency notifications
AI / ML Platform	HTTPS REST	Bidirectional integration with external ML platforms for model training, inference, and monitoring (AWS SageMaker, Azure ML, Google Vertex AI, Anthropic Claude API)
Collaboration Tools	Webhook / REST	Event-driven alerts to Microsoft TE-AMS, Slack, or any webhook endpoint
Asset Discovery Agent	SNMP v3 / WMI	Periodic automated discovery of network-connected assets to identify unregistered devices

10. AI Integration Architecture

10.1 Overview

The E-AMS AI integration layer is designed as a modular, provider-agnostic component that connects the core asset management data platform to one or more AI model providers via the ai_models registry. The architecture ensures that: AI decisions are always traceable; model providers can be swapped without changing application code; and sensitive asset data is handled in accordance with the organisation's data residency and privacy requirements.

10.2 AI Capability Domains

AI Domain	Primary Use Cases	Inputs	Outputs	Storage Table
Predictive Analytics	Hardware failure risk, licence overrun forecast, warranty expiry impact	Asset age, incident history, patch date, vendor EOS dates	Risk score, probability, predicted date	ai_predictions
Anomaly Detection	Unusual audit patterns, after-hours access, bulk exports, privilege escalation	Audit log stream, user behaviour baseline	Anomaly type, severity, supporting evidence	ai_anomalies
Recommendation Engine	Licence consolidation, asset refresh, cost savings, decommission candidates	Utilisation data, cost data, lifecycle data	Recommendations with priority and estimated saving	ai_recommendations
Natural Language Interface	Query asset data in plain English, generate reports, ask compliance questions	User query text, organisation asset data context	Structured query results, narrative responses, generated reports	ai_chat_sessions
Report Narration	Auto-generate executive summaries of standard reports	Report data payload	Plain-language narrative summary	Embedded in report output
Budget Forecasting	Project IT budget requirements for 12–36 months	Historical spend, upcoming expiry/refresh dates	Budget projection with confidence interval	ai_predictions

10.3 Data Privacy and AI Governance

- No raw personally identifiable information (PII) shall be transmitted to external AI model providers without explicit consent and data processing agreement confirmation.
- Data sent to external AI APIs shall be pseudonymised by default: user names replaced with role identifiers, employee IDs with hashed tokens.
- All AI model registrations in the ai_models table shall include a data_sharing_scope field indicating whether the model processes data on-premises, within the organisation's cloud tenant, or via an external third-party API.

- AI models shall not be placed into production without approval by an authorised administrator (is_approved flag in ai_models table).
- Model outputs shall carry a confidence score and shall be presented as recommendations or risk indicators, never as automated decisions that bypass human review.
- The feedback loop (FR-AI-015) shall be used to continuously evaluate model accuracy, with a mandatory review triggered when false positive rate exceeds a configurable threshold.

11. Technical Architecture

11.1 Deployment Architecture

Component	Recommended Technology	Requirement
Application Layer	Python/Django or Node.js/NestJS	Modern maintainable framework; Docker containerisation; Kubernetes orchestration for scalability
Database Layer	PostgreSQL 16+ (primary)	Separate read replica for reporting; automated backups; point-in-time recovery enabled
Cache Layer	Redis 7+	Session caching, notification queuing, frequently accessed dashboard data
Message Queue	RabbitMQ or Apache Kafka	Asynchronous workflow processing, notification dispatch, SIEM log forwarding, AI job scheduling
AI Gateway	FastAPI microservice	Abstraction layer routing AI requests to the appropriate model provider; handles prompt construction and response parsing
Search Engine	Elasticsearch or OpenSearch	Full-text search across asset records and audit logs; powers the natural language query interface
API Gateway	REST API with OpenAPI 3.0 spec	Versioned API; OAuth 2.0 authentication; rate limiting; request/response logging
Load Balancer	NGINX or AWS ALB	Active-passive for production; active-active for horizontal scale deployments
Secrets Manager	HashiCorp Vault or AWS Secrets Manager	Storage of AI API keys, database credentials, SMTP passwords, integration tokens
Monitoring	Prometheus + Grafana	Application metrics, AI model performance metrics, system health dashboards
Log Aggregation	ELK Stack or Loki	Centralised application log collection separate from the immutable audit trail

11.2 Multi-Tenancy and Location Hierarchy

- The system supports both centralised deployment (single instance serving all branches with row-level tenant isolation) and federated deployment (branch instances with data replication to a central instance).

- The centralised model is the recommended default. Branch staff access only assets within their location_id scope by default, with explicit cross-location grants stored in role_permissions.
- Location hierarchy levels (names and number of levels) are configurable per organisation via the Administration Console.

11.3 Open-Source Component Strategy

Where open-source components are used, they shall have an active community, a clear security patch cadence, and an acceptable licence (MIT, Apache 2.0, or BSD preferred). GPL licences shall be evaluated on a case-by-case basis. All third-party component licences shall be documented in a Software Bill of Materials (SBOM) maintained in the system's repository.

12. Glossary

Term	Definition
ACME	Automated Certificate Management Environment — a protocol for automating certificate issuance and renewal
AI	Artificial Intelligence — in this specification, specifically machine learning models used for prediction, anomaly detection, and natural language processing
CMDB	Configuration Management Database — a repository that stores information about IT assets and the relationships between them
E-AMS	E-Asset Management System — the generic name for the system described in this specification
EOS	End of Support — the date on which a vendor ceases providing security patches for a product version
FIDO2	Fast Identity Online 2.0 — a hardware-based passwordless authentication standard
HMAC	Hash-based Message Authentication Code — used for cryptographic signing of audit log entries
HSM	Hardware Security Module — dedicated hardware for cryptographic key management
i18n	Internationalisation — the process of designing software to support multiple languages and locales
KMS	Key Management Service — a service for creating and managing cryptographic keys
LLM	Large Language Model — an AI model trained on large text corpora, used to power the natural-language query interface
MFA	Multi-Factor Authentication — authentication requiring two or more verification factors
OIDC	OpenID Connect — an identity layer on top of OAuth 2.0 for Single Sign-On
RBAC	Role-Based Access Control — a method of controlling access based on the user's assigned role
RPO	Recovery Point Objective — the maximum acceptable data loss in the event of a system failure
RTO	Recovery Time Objective — the maximum acceptable time to restore system operation after a failure

Term	Definition
SAML	Security Assertion Markup Language — an XML-based standard for SSO federation
SBOM	Software Bill of Materials — a list of all third-party components and their licences
SIEM	Security Information and Event Management — platform for real-time security monitoring and correlation
SoD	Segregation of Duties — the principle that no single user should control all steps of a critical process
SSO	Single Sign-On — authentication allowing a single login to access multiple systems
TLS	Transport Layer Security — cryptographic protocol for securing data in transit
UUID	Universally Unique Identifier — a 128-bit identifier guaranteed to be unique across all systems
WCAG	Web Content Accessibility Guidelines — international standards for web accessibility

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